

**A MINI PROJECT TASK REPORT**  
**ON**  
**CUSTOMER SEGMENTATION USING K-**  
**MEANS CLUSTERING**

*Submitted in partial fulfillment of the requirements for the industry internship task suite*

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**DEPARTMENT OF COMPUTER ENGINEERING**  
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## 2. DECLARATION

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I, **Tanishka Rupeshkumar Bellale**, hereby declare that the mini project task report entitled "**Customer Segmentation Using K-Means Clustering**" is an authentic record of the technical work carried out by me as part of the 1-month practical internship training curriculum at **THIRANEX** during the period from May 1, 2026, to May 31, 2026.

The content presented in this report represents true analytical work executed using standard Python scientific libraries. This report is compiled strictly for academic evaluation regarding my B.Tech task completion and has not been submitted elsewhere for any other degree, diploma, or professional certification.

Date: 31/05/2026

Place: Pune

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## 3. ACKNOWLEDGEMENT

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## 4. ABSTRACT

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In modern consumer-centric environments, modern businesses must shift from generalized marketing paradigms to personalized customer engagement. This report details the technical implementation of an unsupervised machine learning task—customer segmentation—using the K-Means clustering algorithm. Utilizing the standard Mall Customers Dataset, the objective is to partition an unlabelled consumer base into explicit cohorts based on two critical economic features: Annual Income and Spending Score. Data preprocessing involved checking for structural alignment, null coordinates, and optimizing feature matrices. The optimal number of target clusters was determined empirically via the Elbow Method, computing the Within-Cluster Sum of Squares (WCSS) across varying parameter thresholds. The execution of the K-Means algorithm established 5 distinct, non-overlapping customer clusters: Premium Customers, Average Customers, Budget Customers, Careful Customers, and Impulsive Customers. Statistical profiling was performed for each cohort to yield actionable business intelligence, proving that unsupervised learning can accurately identify spending behaviors to maximize retail resource allocation.

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## **6. INTRODUCTION**

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The modern retail ecosystem produces extensive transactional and demographic datasets daily. Raw storage, however, does not inherently generate enterprise value. To unlock economic utility, organizations must evaluate user records analytically to extract behavior dynamics. Customer segmentation stands as a cornerstone method in this domain, converting mixed databases into organized target populations.

From a computing perspective, segmentation bypasses the limitations of manual sorting by applying machine learning algorithms. Unsupervised clustering models are uniquely suited for this domain since consumer behavior is fluid and lacks fixed, predefined labels. By examining geometric proximity within feature spaces, clustering discovers natural groupings that human intuition might miss.

## **7. PROJECT OBJECTIVES**

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The primary engineering and operational objectives established for this mini project include:

- To design and deploy a functional machine learning pipeline using Python to achieve automated, unlabelled consumer data parsing.

- To evaluate demographic profiles and isolate the key vectors driving variance in customer purchasing behaviors.
- To mathematically determine the optimal cluster metric using heuristic optimization metrics to prevent arbitrary data division.
- To partition the underlying distribution into 5 distinct behavioral segments based on income structures and internal store spending ratings.
- To compile clear, actionable insights for enterprise application, enabling targeted marketing campaigns and optimized product positioning.

## **8. PROBLEM STATEMENT**

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Traditional marketing strategies often treat all consumers as a single, uniform demographic. This approach results in inefficient budget allocation, untargeted promotional outreach, and subpar customer retention metrics. Because a customer base contains highly diverse income brackets, age variations, and purchasing habits, executing a uniform commercial strategy underperforms. The core technical problem is the lack of explicit target tracking labels in historical retail records. Without an automated, algorithmic system to organize and classify these records based on actual spending attributes, corporate analysts cannot systematically identify valuable or high-churn consumer groups.

## **9. SCOPE OF THE PROJECT**

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The scope of this mini project centers on analyzing the numerical and behavioral vectors within the Mall Customers Dataset using unsupervised machine learning. The analysis focuses primarily on evaluating the mathematical correlation between two key features: Annual Income and Spending Score. The computational pipeline covers data acquisition, preprocessing, spatial configuration checks, iterative model training, cluster boundaries mapping, and business intelligence profile generation. This project excludes real-time streaming pipelines, dynamic online model updates, or deployment into production cloud instances. Instead, it serves as a robust analytical framework for offline retail strategy formulation.

## 10. TOOLS AND TECHNOLOGIES USED

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The technological stack selected for this machine learning task relies entirely on the Python scientific ecosystem, ensuring optimal memory efficiency and modular script assembly. The specific utilities are detailed below:

- **Python:** The underlying high-level object-oriented programming language chosen for its extensive numerical packages and clean processing syntax.
- **Jupyter Notebook:** The interactive development environment used for rapid model prototyping, exploratory data analysis, and inline data visualization.
- **Pandas:** Used for structure handling, reading raw data, tracking data frames, and executing filtering tasks.
- **NumPy:** Used to manage high-dimensional vector grids, multi-dimensional array transformations, and low-level algebraic calculations.
- **Matplotlib & Seaborn:** Used to construct high-resolution statistical visualizations, including 2D scatter plots, distribution plots, and line charts.
- **Scikit-learn:** The core machine learning library used to import, configure, and execute the *K-Means* optimization module and evaluate clustering metrics.

## 11. DATASET DESCRIPTION

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The empirical research utilized the standardized *Mall Customers Dataset*, capturing core demographic and behavioral indicators for a collection of retail consumers. The dataset contains 200 individual customer records across 5 structural features:

| Feature Title  | Data Typology        | Mathematical Nature | Analytical Function / Description                                                    |
|----------------|----------------------|---------------------|--------------------------------------------------------------------------------------|
| CustomerID     | Integer (Numeric)    | Discrete / Nominal  | Unique database key assigned to identify individual consumer records.                |
| Gender         | String (Categorical) | Nominal Binary      | Identifies individual gender identification (Male / Female).                         |
| Age            | Integer (Numeric)    | Continuous Ratio    | Chronological age of the customer recorded in years.                                 |
| Annual Income  | Integer (Numeric)    | Continuous Ratio    | Annual self-reported earnings of the consumer scaled in thousands of dollars (\$k).  |
| Spending Score | Integer (Numeric)    | Discrete Interval   | Internal index value (1 to 100) assigned by the facility based on purchasing habits. |

## 12. DATA PREPROCESSING

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Before training the machine learning model, the raw dataset underwent comprehensive preprocessing to ensure data integrity and model stability. The processing sequence followed three main phases:

**1. Structural Verification:** The shape of the matrix was verified using `df.shape` to ensure 200 rows and 5 core columns were correctly loaded. Data types were confirmed via `df.info()`.

**2. Missing Value Audit:** Null values can introduce bias and cause runtime execution failures in Euclidean distance calculations. An explicit missing data scan was executed using `df.isnull().sum()`, confirming zero missing values across all features, which bypassed the need for imputation.

**3. Feature Extraction:** To segment customers by economic behavior, the relevant feature vectors were isolated. The `CustomerID`, `Gender`, and `Age` dimensions were filtered out, extracting `Annual Income` (Column Index 3) and `Spending Score` (Column Index 4) into a dedicated processing array  $X = df.iloc[:, [3, 4]].values$ .

## 13. EXPLORATORY DATA ANALYSIS

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Exploratory Data Analysis (EDA) was performed to understand the underlying data distribution before applying clustering algorithms. The analysis revealed a balanced distribution across the primary features. Age ranges span from 18 to 70 years, with a high density of active mall consumers concentrated between 30 and 45 years old.

The statistical spread of Annual Income ranges from \$15k to \$137k, with the majority of samples clustered between \$60k and \$75k. Similarly, the Spending Score spans from 1 to 100, showing a notable concentration around the median value of 50. A baseline scatter plot mapping raw income against spending scores revealed several natural visual groupings, confirming that the dataset is well-suited for unsupervised spatial clustering.

## 14. K-MEANS CLUSTERING ALGORITHM

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The **K-Means** algorithm is a fundamental unsupervised machine learning technique used to partition an unlabelled dataset into  $k$  distinct, non-overlapping groups. The algorithm treats each data point as a vector in a multi-dimensional space, iteratively organizing samples based on geometric proximity.

The underlying optimization relies on minimizing spatial variance. The distance between a data point  $x_i$  and a cluster centroid  $\mu_j$  is computed using the standard Euclidean distance metric:

$$d(x_i, \mu_j) = \sqrt{\sum_{k=1}^n (x_{ik} - \mu_{jk})^2}$$

The primary objective of the algorithm is to minimize the total sum of squared distances within each cluster. This objective function is mathematically formulated as:

$$J = \sum_{j=1}^k \sum_{i=1}^m \|x_i^{(j)} - \mu_j\|^2$$

The iterative execution follows a structured multi-step optimization framework:

- **Initialization:** The model selects  $k$  initial points to serve as the starting cluster centroids. This implementation utilizes the **k-means++** initialization strategy to place initial centroids far apart, accelerating convergence and avoiding sub-optimal local minima.
- **Assignment Step:** The algorithm calculates the Euclidean distance from every individual data point to all  $k$  centroids, assigning each point to its nearest cluster.



- **Update Step:** The algorithm recalculates the position of each centroid by taking the mathematical mean of all data points currently assigned to that cluster:  $\mu_j = (1 / |S_j|) \sum_{x_i \in S_j} x_i$ .
- **Convergence:** Steps 2 and 3 repeat iteratively until the centroid coordinates stabilize, meaning no data points change cluster assignments, or the maximum iteration threshold is met.

## 15. ELBOW METHOD EXPLANATION

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A fundamental challenge in unsupervised learning is determining the optimal number of clusters  $k$  when target labels are missing. Selecting an arbitrary  $k$  can result in underfitting or overfitting. To solve this, the Elbow Method was used to determine the optimal cluster count empirically.

The Elbow Method evaluates the Within-Cluster Sum of Squares (WCSS) across a range of cluster counts. WCSS measures the compactness of the clusters, calculated as:

$$WCSS = \sum_{j=1}^k \sum_{x_i \in S_j} dist(x_i, \mu_j)^2$$

As  $k$  increases, individual WCSS values naturally decrease because data points sit closer to their respective centroids. In an extreme case where  $k = N$  (the total number of data points), the WCSS drops to zero. The optimal configuration is found by plotting WCSS against a range of  $k$  values (typically 1 to 10) and identifying the distinct "elbow point." This point represents the threshold where increasing the cluster count yields diminishing returns in variance reduction.

## 16. MODEL IMPLEMENTATION

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The complete customer segmentation pipeline was implemented sequentially in a Python environment. The following structured sequence outlines the core execution steps:

**1. Running the Elbow Method:** An explicit loop was written to initialize and fit the KMeans model for cluster sizes ranging from 1 to 10. The WCSS metrics were captured via the `inertia_` attribute, as shown in the tracking sequence below:

- For  $k = 1$ : WCSS  $\approx$  269,981.28
- For  $k = 2$ : WCSS  $\approx$  181,363.50
- For  $k = 3$ : WCSS  $\approx$  106,111.01

- For  $k = 4$ : WCSS  $\approx 73,679.34$
- For  $k = 5$ : WCSS  $\approx 44,448.45$
- For  $k = 6$ : WCSS  $\approx 37,233.81$

Plotting these values revealed a clear elbow at  $k = 5$ , indicating the optimal number of segments for the dataset.

**2. Final Model Execution:** With the optimal cluster count established, the final model was initialized and executed using the following parameters:

```
kmeans = KMeans(n_clusters=5, init='k-means++', random_state=42)
y_kmeans = kmeans.fit_predict(X)
```

This assigned each of the 200 data points a cluster index label from 0 to 4, which was then appended back to the core dataframe for behavioral profiling.

## 17. CLUSTER ANALYSIS AND INTERPRETATION

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Analyzing the resulting spatial coordinate boundaries revealed 5 distinct customer profiles based on income distributions and store spending metrics:

- **Cluster 1 (Index 0) - Average Customers:** Positioned directly in the center of the distribution. These consumers display moderate annual income brackets (\$40k to \$65k) alongside average spending metrics (40 to 60). They represent a stable, predictable middle-class consumer cohort.
- **Cluster 2 (Index 1) - Premium Customers:** Characterized by high annual earnings (\$70k to \$140k) paired with high spending scores (70 to 100). This group represents a highly profitable target segment for premium product placements.
- **Cluster 3 (Index 2) - Careful Customers:** Characterized by high annual earnings (\$70k to \$140k) but low spending scores (1 to 40). These consumers are financially secure but highly deliberate or conservative in their retail purchasing behaviors.
- **Cluster 4 (Index 3) - Impulsive Customers:** Characterized by low annual income levels (\$15k to \$40k) but exceptionally high spending scores (60 to 100). This group demonstrates frequent, spontaneous purchasing patterns that outpace their baseline income.
- **Cluster 5 (Index 4) - Budget Customers:** Positioned in the lower-left region of the spatial distribution. These consumers have low annual income levels (\$15k to \$40k) and low

spending scores (1 to 40), aligning their retail expenditures with their constrained budget limits.

## 18. RESULTS AND FINDINGS

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The mathematical evaluation confirmed that the 5 isolated clusters are well-separated and stable, with distinct geometric centroids. The specific centroid coordinates are summarized below:

| Assigned Cluster Label | Cohort Title        | Centroid Annual Income (\$k) | Centroid Spending Score (1-100) | Sample Volume |
|------------------------|---------------------|------------------------------|---------------------------------|---------------|
| Cluster 0              | Average Customers   | \$55.29k                     | 49.51                           | 81 counts     |
| Cluster 1              | Premium Customers   | \$86.53k                     | 82.12                           | 39 counts     |
| Cluster 2              | Careful Customers   | \$88.20k                     | 17.11                           | 35 counts     |
| Cluster 3              | Impulsive Customers | \$25.72k                     | 79.36                           | 22 counts     |
| Cluster 4              | Budget Customers    | \$26.30k                     | 20.91                           | 23 counts     |

The Average Customers group forms the largest cohort (81 out of 200), establishing a solid baseline consumer profile. The Premium and Careful cohorts show similar high income levels but contrasting spending behaviors, underscoring the limitations of using income alone as a segmentation feature.

## 19. BUSINESS INSIGHTS

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The mathematical boundaries discovered by the model translate directly into actionable strategies for retail management and marketing teams:

- **Premium Customization:** Marketing teams should target the Premium Customers cohort with exclusive product launches, personalized loyalty rewards, and VIP event access, maximizing revenue from high-value consumers.

- **Conversion Campaigns:** The Careful Customers group presents a major growth opportunity. Because they have high disposable income but low spending scores, they can be targeted with personalized promotions, product demonstrations, and value-focused campaigns to encourage purchasing.
- **Engagement Maintenance:** The large Average Customers group requires steady engagement through standard seasonal discounts and loyalty programs to maintain retention without eroding profit margins.
- **Tactical Promotions:** Impulsive Customers respond well to limited-time offers, flash sales, and point-of-purchase displays that encourage spontaneous buying.

## 20. ADVANTAGES OF CUSTOMER SEGMENTATION

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Implementing algorithmic customer segmentation provides several key operational advantages:

- **Optimized Resource Allocation:** Enables marketing teams to allocate budgets efficiently by focusing high-cost campaigns on high-yield customer segments.
- **Enhanced Personalization:** Supports the creation of tailored promotional messaging that aligns with the specific financial profile and spending habits of each cohort.
- **Improved Product Positioning:** Guides inventory management by aligning stock levels and product categories with the dominant customer profiles of specific retail locations.
- **Proactive Churn Mitigation:** Allows analysts to detect shifts in cluster assignments early, signaling potential customer dissatisfaction or attrition risks.

## 21. CHALLENGES FACED

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Several technical and analytical challenges were addressed during the execution of this internship task. First, unsupervised algorithms can be sensitive to scale imbalances. Although the features in this dataset shared comparable ranges, ensuring stable variance required careful evaluation. Second, determining the exact location of the elbow point can be subjective when the WCSS curve decreases smoothly. This was resolved by computing variance differentials across successive step changes. Finally, the model relies on static demographic features; incorporating dynamic, time-series transactional data would require more complex sequence-based architectures.

## 22. LEARNING OUTCOMES

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The completion of this practical task provided significant technical and professional learning benefits:

- Gained hands-on experience building end-to-end machine learning pipelines using Python scientific libraries.
- Developed a deep understanding of unsupervised learning mathematics, specifically vector distances and centroid optimization.
- Mastered empirical hyperparameter tuning by implementing the WCSS validation framework to find optimal cluster counts.
- Learned to interpret complex visual distributions and translate data clusters into practical business strategies.

## 23. CONCLUSION

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This mini project successfully demonstrates the application of the K-Means clustering algorithm for customer segmentation on the Mall Customers Dataset. By analyzing the relationship between annual income and spending behaviors, the model extracted 5 distinct consumer cohorts without relying on predefined labels. The empirical determination of  $k = 5$  using the Elbow Method provided a solid mathematical foundation for the segmentation framework. The clear visual separation between the clusters proves that unsupervised learning can uncover meaningful behavioral patterns in complex retail data, providing businesses with a reliable tool to optimize marketing efficiency and customer engagement.

## 24. FUTURE SCOPE

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The current analytical framework can be expanded in several key areas. First, incorporating additional feature dimensions—such as online browsing history, purchase frequency, and geographical location—would create a more comprehensive, multi-dimensional segmentation model. Second, implementing density-based clustering algorithms, such as DBSCAN, could help identify irregular or non-spherical customer distributions and isolate outliers. Finally, integrating this pipeline with automated streaming platforms would enable real-time segment updates, allowing marketing systems to adapt dynamically to changing consumer behaviors.

## 25. REFERENCES

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